TERTIARY INFOTECH ACADEMY PTE LTD
C239

ACTIVITY A3
Select and Apply Quality Tools

Choose quality tools that fit the question, data type and customer segment.

Participant scenario and completion checklist

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Scenario

LPC has 60 recent complaints from three customer segments. The descriptions are inconsistent and teams jump directly to favourite solutions.

Organisation: LionCity Precision Components Pte Ltd (LPC). A Singapore SME with 85 employees producing precision aluminium components for commercial equipment.

Your task

Use check sheets, Pareto analysis and cause-and-effect analysis to prioritise LPC's customer complaints.

Expected output: A coded complaint check sheet, Pareto chart and evidence-led cause hypothesis.

Decision questions

1. Which tool answers 'what happens most often' and which explores 'why might it happen'?

2. Do the priorities change by customer segment or product family?

3. Which possible cause needs verification before it can be treated as root cause?

Completion checklist

Done	Check	Evidence/reference
[ ]	Code raw complaint descriptions into consistent categories.	
[ ]	Validate category definitions on a sample with a partner.	
[ ]	Build a frequency table and Pareto chart with cumulative percentage.	
[ ]	Stratify the vital categories by customer segment.	
[ ]	Build a fishbone hypothesis and specify evidence needed to test each major branch.	
[ ]	All conclusions are traceable to evidence; assumptions are labelled.	
[ ]	A peer has challenged one key decision and the response is recorded.	
[ ]	The workbook passes the acceptance test below.	

Acceptance test

The Pareto chart reconciles to the complaint total and the selected cause hypothesis is explicitly labelled for verification.

Submission and debrief

Submit the completed Excel workbook using the trainer's naming convention. Be ready to explain one evidence-based decision, one uncertainty, and one improvement you would make with better data.